Chemosphere 286 (2022) 131565

Contents lists available at ScienceDirect

Chemosphere

journal homepage: www.elsevier.com/locate/chemosphere

Extensive production of hydroxyl radicals during oxygenation of anoxic paddy soils: Implications to imidacloprid degradation

Wenchao Wang, Danyu Huang, Dixiang Wang, Mengxi Tan, Mengyuan Geng, Changyin Zhu, Ning Chen *, Dongmei Zhou

State Key Laboratory of Pollution Control and Resource Reuse, School of the Environment, Nanjing University, Nanjing, 210023, Jiangsu Province, PR China

ARTICLE INFO

Handling Editor: Yongmei Li

Keywords:
Hydroxyl radicals
Paddy soil
Iron species
Imidacloprid
Degradation

ABSTRACT

Hydroxyl radical (•OH) plays a critical role in driving organic pollutants degradation during redox fluctuations. Such processes have been frequently investigated in sedimentary environments, but rarely referred to the agricultural fields, such as paddy soils with frequent occurrence of redox fluctuations. Our findings demonstrated that extensive •OH (40.3–1061.4 μmol kg^{-1}) was produced during oxygenation of anoxic paddy slurries under circumstance conditions. Wet chemical sequential extractions, Mössbauer spectra, and X-ray photoelectron spectroscopy characterizations collectively corroborated that 0.5 M HCl-extracted Fe(II) (i.e., surface-bound Fe and Fe in low-crystalline minerals) contributed to more •OH production than aqueous Fe^{2+}. The produced •OH can efficiently induce the oxidative transformation of organic carbon and the degradation of imidacloprid (IMP), which in turn produced the by-products, such as IMP-urea, IMP-olefin, and 6-chloronicontinic acid, via •OH-attacking mechanisms. Quenching experiments showed that hydrogen peroxide (H_2O_2) was the important intermediate for •OH formation via Haber-Weiss mechanisms during oxygenation processes. These findings indicate that abundant •OH can be produced during the redox fluctuations of paddy soil, which might be of great significance to predict the removal of organic contaminants and the mineralization of organic carbon in paddy fields.

1. Introduction

As one of the most powerful oxidants in natural environment, hydroxyl radicals (•OH, $E^0 = 2.7$ V) can serve as the important initiators in chemical oxidation processes and mediate the degradation or transformation of organic pollutants (Gligorovski et al., 2015; Tong et al., 2016; Chen et al., 2018b, 2019a, 2019a; Wang et al., 2021), organic matters (Yu and Kuzyakov, 2021; Zeng et al., 2020), and nutrients (Gligorovski et al., 2015). In recent years, increasing studies have reported that •OH can be produced from the oxygenation of reduced iron (Fe) species (e.g., aqueous Fe^{2+}, ligand complexed Fe(II) and Fe (II)-bearing minerals) (Garg et al., 2018; Han et al., 2020; Santana-Casiano et al., 2005; Zhang et al., 2018a; Zhang and Yuan, 2017), and reduced organic matters by oxygen (O_2) (Page et al., 2012; Qin et al., 2016), which also showed the high efficiency of •OH-mediated degradation of tetracyclines (Tong et al., 2016), phthalate esters (Chen et al., 2019a), trichloroethylene (Cheng et al., 2020; Liu et al., 2017), and the decomposition of organic matters (Zeng et al., 2020; Waggoner et al., 2015; Zhang et al., 2020). Thus, more attention has been paid to •OH production through oxygenation of anoxic sediments or surface waters in recent years (Page et al., 2013; Tong et al., 2016; Zhang et al., 2020). Compared with the sedimentary systems, limited studies have investigated the •OH formation in agricultural soils, especially the paddy soils with frequent occurrence of redox fluctuations.

The physicochemical properties and biogeochemical processes of paddy soils are largely different to that of sediments (Kögel knabner et al., 2010). Redox condition of paddy soil changes frequently due to the water management during rice growth. The frequent occurrence of redox fluctuations affects the structure and function of microbial communities of paddy soil, which highly impacts the short-term biogeochemical processes, such as Fe phases oxidation and reduction, and organic matter transformation (Hanke et al., 2013; Kögel knabner et al., 2010; Xu et al., 2019). Vogelsang et al. (2016) proved that redox fluctuations can induce the accumulation of organic matter, poorly-crystalline Fe oxides in the topsoil layers of paddy soil. Rapid redox oscillations may also intensify the mineral weathering and

* Corresponding author.
E-mail address: chenning@nju.edu.cn (N. Chen).

https://doi.org/10.1016/j.chemosphere.2021.131565
Received 11 May 2021; Received in revised form 22 June 2021; Accepted 13 July 2021
Available online 15 July 2021
0045-6535/© 2021 Elsevier Ltd. All rights reserved.

leaching processes (Kögel knabner et al., 2010; Zhao et al., 2021), which further alters the crystallinity and reactivity of Fe phase in paddy soil (Thompson et al., 2006; Winkler et al., 2018), and probably •OH formation. Paddy soils cultivated from different soil types might contain discrepant amounts of the low-crystalline (e.g., ferrihydrite) and high-crystalline Fe phases (e.g., goethite) (Wang et al., 2019b), which in turn may impact the processes of redox fluctuations and •OH production. Besides, paddy soils are easily subjected to the agricultural tillage, cropping and rainfall, which causes the quick penetration of oxygen into soil, potentially leading to the rapid oxidation of reduced substances (Watanabe et al., 2021). Hence, these collective factors associated with different types of paddy soils might significantly affected the Fe species transformation and •OH production during redox fluctuations.

Previous studies have proved that Fe (II) was mainly responsible for •OH production (Page et al., 2013; Tong et al., 2016; Zhang et al., 2020), but not all Fe(II) species exerts the positive effects (Tong et al., 2016). Xie et al. (2020) demonstrated that the surface-adsorbed and mineral structural Fe(II) might serve as the major contributors to •OH production in oxygenation of anoxic sedimentary slurries. However, the physicochemical properties of paddy soil (especially the content and crystallinity of Fe phases) cultivated from different types of soils were largely different to each other, which might induce the discrepant contributions of Fe species to •OH production in different paddy soils. Therefore, the active Fe(II) species for •OH production need to be identified in different paddy soils and the underlying mechanisms should be systematically examined. Besides, the role of •OH in driving organic pollutants degradation and organic matters transformation need to be explored concurrently, which may be of great significance to predict the fate of organic pollutants and the transformation of organic carbon induced by abiotic processes.

The main purposes of this study are to explore the processes and mechanisms of •OH production during oxygenation of anoxic paddy slurries and identify the contributions of different Fe species to •OH production. Wet chemical extractions were applied to quantify the dynamic changes of Fe species, which were also concomitantly characterized by state-of-art spectroscopies (e.g., Mössbauer spectroscopy, X-ray photoelectron spectroscopy (XPS), Fourier transform infrared spectrometer (FTIR)). For comparison, paddy soil from different depths (0–20 cm and 40–60 cm) were collected at five sampling sites (Chengdu, Changsha, Jiangdu, Yingtan, and Changchun) in the main rice-production region of China. To investigate the organic pollutants degradation by •OH, imidacloprid (IMP) was chosen as a model organic pollutant due to its wide distribution in paddy soils (Sharma and Singh, 2014), and its degradation products were identified by liquid chromatography quadrupole-time-of-flight mass spectrometry (LC-QTOF-MS). The •OH-induced oxidative transformation of organic carbon (OC) was examined concomitantly.

2. Materials and methods

2.1. Chemicals

Benzoic acid (BA, 98%) and hydroxylammonium hydrochloride ($NH_2OH{\cdot}HCl$, 99%) were purchased from Sinopharm Chemical Reagent Co, Ltd, China. Sodium dithionite ($Na_2S_2O_4$, 88%), 1, 10-phenanthroline ($C_{12}H_8N_2$, 99%), IMP (98%), catalase (40,000 units mg^{-1}), ammonium acetate ($C_2H_7NO_2$, 99%), glacial acetic acid ($C_2H_4O_2$, 99.7%), sodium pyrophosphate ($Na_4P_2O_7{\cdot}10H_2O$, 99%), hydrochloric acid (HCl, 36.5%), 2, 2-bipyridine, (2, 2-BPY, 98%), nitro blue tetrazolium chloride (NBT, 99%), methanol (CH_3OH, 99.7%), and acetonitrile (C_2H_3N, 99.7%) were obtained from Sigma-Aldrich. All chemicals were used as received without purification. All solutions were prepared using ultrapure water (18.2 MΩ cm^{-1}) from a Synergy UV ultrapure water system with Millipak-40 filter unit (Millipore Corporation Merck KGaA, Darmstadt, Ge).

2.2. Sampling of paddy soils and anoxic incubation

The sampling sites were located at Chengdu (CD, 118°56′E, 32° 6’°N) in Sichuan province, Yingtan (YT, 117°12′ E, 32°6′ N) in Jiangxi province, Changchun (CC, 125.35°E, 43.88°N) in Jilin province, Jiangdu (JD, 32°19′ E, 119°28′ N) in Jiangsu province, and Changsha (CS, 113.15°E, 28.41°N) in Hunan province, which were classified as Entisol, Ultisol, Mollisol, and Inceptisol (JD and CS) based on US soil taxonomy. Rice was harvested twice per year in these paddy sites, except for CC (once per year). Soil samples were collected from the suburban paddy sites at depth of 0–20 cm or 40–60 cm ($CD_{0\text{-}20}$ and $CD_{40\text{-}60}$ represent the soil depth of 0–20 cm and 40–60 cm, respectively) at the end of March in 2019. Soil samples were air dried, homogenized and passed through the 100-mesh sieve. The details for the measurements of physicochemical properties of soil samples are concluded in Text S1 and Table S1.

In anoxic incubation, 20 g soil and 50 mL deionized water (degassed with 99.9% N_2 for 120 min) were well mixed in 120 mL serum bottles in an anoxic glovebox (Braun Co, Germany). Then, these bottles were sealed with thick butyl rubbers and placed on reciprocating shaker (250 rpm, 25 °C) to re-establish the activities of microorganisms. The Eh and pH were measured during the incubation, and oxygenation experiments were performed till the Eh and pH of incubated slurries were metastable after 30 d (Figure S1).

2.3. Batch oxygenation experiments

Batch oxygenation experiments were conducted in triplicate with the sacrificed mode. In the typical experiments, 10 mL homogeneous suspension (containing ~0.31 g dry soil) was added into 20 mL glass bottles under anoxic conditions. Then, these bottles were exposed to air to initiate oxygenation and then vibrated at 200 rpm under dark conditions (Tong et al., 2016). At predetermined time intervals, 1 mL of dispersed suspension was withdrawn for the measurements of different Fe species.

For the determination of •OH, 1 mL 200 mM BA (•OH probe) and 9 mL homogeneous suspension were well-mixed in 20 mL glass bottles and then oxygenated in the same procedure. The anoxic controls were also performed in the presence of 99.9% N_2 in headspace. At predetermined time intervals, 1 mL of dispersed suspension was withdrawn from the bottles and immediately mixed with 1 mL methanol (MeOH) to quench the reactions. After centrifugation and filtration of 0.22 μm membrane, *p*-hydroxybenzoic acid (*p*-HBA) concentrations were analyzed by HPLC. To screen the effects of microorganisms, a paralleled set of incubated bottles were sacrificed and performed with γ irradiation ($9 \times 10^{\wedge 6}$Ci, Co_{60}) at room temperature and then oxygenated as above.

2.4. IMP degradation

To examine the IMP degradation by •OH, 20 g soil and 50 mL 20 μM IMP solution were added into a 120 mL serum bottle under anoxic conditions and then incubated for 20 days. After incubation, the residual IMP in soil slurries were extracted and analyzed. Then, 10 mL homogeneous suspension was exposed to air for oxygenation as above. At predetermined time intervals, parallel tubes were sacrificed for IMP analyses. For residual IMP extraction, 12.8 mL MeOH and 3.2 mL 1 mM HCl were added concurrently into the reaction tubes, and shaken for 2 h (Chen et al., 2019b; Lesueur et al., 2008; Sharma and Singh, 2014). Then, 1 mL homogeneous suspension was withdrawn and passed through 0.22 μm filter membrane for HPLC analyses. The recovery of the extraction method was 88 ± 5%. All experiments were performed in duplicates with standard deviations reported in results.

2.5. Analytical methods

2.5.1. Sequential extraction of Fe species

Two sets of sequential extractions were conducted in parallel to extract the Fe(II) and total Fe species based on the established methods

(Voelz et al., 2019; Xie et al., 2020). In the first set, soil suspensions were sequentially extracted by the 5 mL ultrapure water (degassed with Ar for 2 h) and 0.5 M HCl for aqueous Fe^{2+}, and complexed or surface-adsorbed Fe(II) and Fe(II) in low-crystalline minerals (e.g., carbonates and green rusts), respectively (Chen et al., 2018a; Voelz et al., 2019; Williams and Scherer, 2001). In the second set, soil suspensions were sequentially extracted by the 5 mL 0.1 M $Na_4P_2O_7$, 0.25 M $NH_2OH{\cdot}HCl$ and 60 mM $Na_2S_2O_4$ to extract OM-complexed Fe, low-crystalline Fe, and high-crystalline Fe, respectively (Poulton and Canfield, 2005; Xie et al., 2020). After the extraction and centrifugation, the filtrates was preserved in 0.1 M HCl in glove box until analyses. The Fe(II) and total Fe (TFe) reduced by 10% hydroxylamine hydrochloride were measured at 560 nm with a UV–Vis spectrometer (Shimadzu UV-2550) using the phenanthroline method (Amonette and Tampleton, 1998). (Details of the extraction procedures are included in Text S2).

2.6. Mössbauer spectroscopy characterization

To further characterize the Fe(II) species during oxygenation, soil particles before and after oxygenation were characterized by Mössbauer spectroscopy. After anoxic incubation, soil suspension was withdrawn and centrifuged under anoxic condition. Then, soil samples were freezing-dried, ground, and preserved with O_2-impermeable Kapton tape before analyses. The Mössbauer spectra was collected using a conventional spectrometer (Germany, Wissel MS-500) in transmission geometry with constant acceleration mode at 13 K. $A^{57}Co(Rh)$ source with activity of 25 mCi was used. The velocity calibration was done with a room temperature α-Fe absorber. The spectra were fitted by the software Recoil using Lorentzian multiple analyses, with parameters listed in Table S2.

2.7. XPS analyses

XPS spectroscopy was applied to characterize the surface Fe(II) and OC species during oxygenation. All XPS samples were prepared same as the Mössbauer characterization. The XPS measurements were carried out on an ESCALAB 250Xi spectrometer (Thermo Scientific, USA) equipped with a pass energy of 30 eV with a power of 100 W (10 kV and 10 mA) and a mono-chromatized AlKα X-ray ($h\nu = 1486.65$ eV) source. All samples were analyzed under pressure of less than $1.0 \times 10^{\hat{}-9}$ Pa. Spectra were acquired through the Avantage software (Version 5.979) with a step of 0.05 eV. To obtain the oxidation status of surface sites of Fe oxides and targeted contaminants, the narrow scan spectra for Fe 2*p* and C 1*s* were acquired. The carbon 1*s* electron binding energy corresponding to graphitic carbon at 284.8 eV was used as reference for calibration. Narrow scan spectra of Fe2$p_{3/2}$ and C1*s* were collected as binding energy forms and fitted using a nonlinear least-squares curve-fitting program (XPSPEAK41 software).

2.8. Other analyses

The *p*-HBA: *p*-HBA concentration was determined using the HPLC (Shimadzu LC-2030 Plus). Mobile phases were the mixture of 35% methanol and 0.1% phosphoric acid, with retention time at 9.7 min. The column oven temperature was 40 °C, and the wavelength was set at 254 nm. The concentration of *p*-HBA was used to estimate the cumulative $^{\bullet}OH$ with a conversion factor of 5.87 (Mopper and Zhou, 1990). The mobile phases of IMP were the mixture of 30% acetonitrile and 70% ultrapure water, with the column oven temperature set at 30 °C. The wavelength was set at 270 nm with the retention time at 5.8 min (Bonmatin et al., 2003; Wang et al., 2020). The degradation products of IMP were qualitatively monitored by LC-TOF-MS/MS with an AB ×500R LC-QTOF mass spectrometer (AB Sciex, USA). In brief, chromatographic separation of transformed compounds was performed at 30 °C with a LC-18 column (2.1 × 100 mm, 2 μm). The mobile phase was a mixture of 80% 0.01 M formic acid and 20% acetonitrile at 0.2 mL min^{-1}. Data acquisition was conducted in a positive ESI mode with the *m/z* scan range of 50–1000. The optimized MS/MS parameters were set as: source temperature, 120 °C; capillary voltage, 3.2 kV; cone voltage, 30 V; nebulizing gas flow, 3 L min^{-1}; nebulizer pressure, 160 kPa.

3. Results and discussion

3.1. $^{\bullet}OH$ production during oxygenation of anoxic paddy soil

Negligible $^{\bullet}OH$ accumulated under anoxic conditions, whereas extensive $^{\bullet}OH$ accumulation was observed during oxygenation process (Fig. 1a and S2). For $CD_{0\text{-}20}$ slurry, $^{\bullet}OH$ accumulated rapidly within the initial 2 h ($k_{obs} = 79.43 \pm 0.01$ μM h^{-1}, the $^{\bullet}OH$ accumulation within the initial 2 h well followed the zero-order kinetics) and then slowly afterwards, producing 206 ± 38 μM $^{\bullet}OH$ after 12 h oxygenation. Compared with $CD_{0\text{-}20}$ slurry, $^{\bullet}OH$ accumulation rate of $JD_{0\text{-}20}$ (90.18 ± 16.62 μM h^{-1}) and $CS_{0\text{-}20}$ (92.30 ± 15.18 μM h^{-1}) slurries were much higher, which produced 310 ± 13 μM and 300 ± 36 μM $^{\bullet}OH$ after 12 h oxygenation, respectively. However, the $CC_{0\text{-}20}$ (111.33 ± 16.37 μM h^{-1}) and $YT_{0\text{-}20}$ (106.35 ± 3.11 μM h^{-1}) showed the highest $^{\bullet}OH$ accumulation rate among these paddy slurries, inducing 322 ± 4 and 305 ± 3 μM $^{\bullet}OH$ accumulation after 12 h oxygenation, respectively. These results indicated that the $^{\bullet}OH$ production was largely different in these paddy slurries, which was probably attributed to discrepancies of Fe(II) oxidation processes. Compared to the $CD_{0\text{-}20}$, $JD_{0\text{-}20}$ and $YT_{0\text{-}20}$ slurries, $^{\bullet}OH$ accumulation in $CD_{40\text{-}60}$, $JD_{40\text{-}60}$ and $YT_{40\text{-}60}$ was limited (Fig. 1b), giving values of 3.61 ± 0.08, 3.63 ± 0.49 and 3.56 ± 0.27 μM after 12 h, respectively. The decreased $^{\bullet}OH$ accumulation may suggest that active Fe(II) species significantly declined in deep soils. Oxygenation of these paddy slurries can produce 669.5–1046.5 μmol $^{\bullet}OH$ kg^{-1} dry soil (at water soil ratio of 2.5:1), which is a bit higher than that of sediments (2–670 μmol $^{\bullet}OH$ kg^{-1} dry sediment) (Tong et al., 2016), but magnitudes higher than that of surface waters (Page et al., 2013). Hence, redox fluctuations of paddy soil can produce large amounts of $^{\bullet}OH$, which was highly ignored in previous studies.

To examine the role of microorganisms in driving $^{\bullet}OH$ formation, γ irradiation was performed after anoxic incubation due to its high efficiency in killing microorganisms (Wang et al., 2019a). Results showed sterilization of paddy soil negligibly affected the $^{\bullet}OH$ accumulation during oxygenation (Figure S3), which indicated the oxygen-mediated abiotic processes were mainly responsible for $^{\bullet}OH$ formation during oxygenation of anoxic paddy slurries.

3.2. Exploring the active Fe species for $^{\bullet}OH$ production

The initial Fe^{2+} concentration in the $CD_{0\text{-}20}$, $JD_{0\text{-}20}$, $CC_{0\text{-}20}$ and $CS_{0\text{-}20}$ slurries were 0.517 ± 0.037, 0.522 ± 0.011, 0.632 ± 0.035 and 0.518 ± 0.006 mg L^{-1}, respectively (Fig. 2a), which indicated more aqueous Fe^{2+} presented in the $CC_{0\text{-}20}$ slurry than that of other paddy slurries. The Fe^{2+} decreased rapidly within the initial 2 h and slowly afterwards. After 8 h oxygenation, the oxidized Fe^{2+} content in the $CD_{0\text{-}20}$, $JD_{0\text{-}20}$, $CC_{0\text{-}20}$ and $CS_{0\text{-}20}$ slurries were 0.095 ± 0.04, 0.101 ± 0.015, 0.183 ± 0.035 and 0.096 ± 0.055 mg L^{-1} respectively, suggesting more oxidation of Fe^{2+} occurred in the $CC_{0\text{-}20}$ slurry.

The initial Fe^{2+} content in the $CD_{40\text{-}60}$ slurry (0.432 ± 0.031 mg L^{-1}) was lower than that of the $CD_{0\text{-}20}$ slurry. After 8 h oxygenation, the consumed Fe^{2+} content in the $JD_{40\text{-}60}$ slurry was 0.061 ± 0.001 mg L^{-1} lower than that of $JD_{0\text{-}20}$ slurry, which indicated that the decreased Fe^{2+} content was responsible for the declined $^{\bullet}OH$ formation in the $JD_{40\text{-}60}$ slurry.

The 0.5 M HCl extracted Fe (0.5 M HCl–Fe) was regarded as the indication of redox status of soils and sediments (Heron et al., 1994). Hence, changes of 0.5 M HCl–Fe(II) was further investigated during oxygenation processes. The data points within the initial 2 h were fitted well with the zero order kinetics (Fig. 3b), which showed that the k_{obs} of 0.5 M HCl–Fe(II) oxidation in the $JD_{0\text{-}20}$, $YT_{0\text{-}20}$, $CC_{0\text{-}20}$ and $CS_{0\text{-}20}$

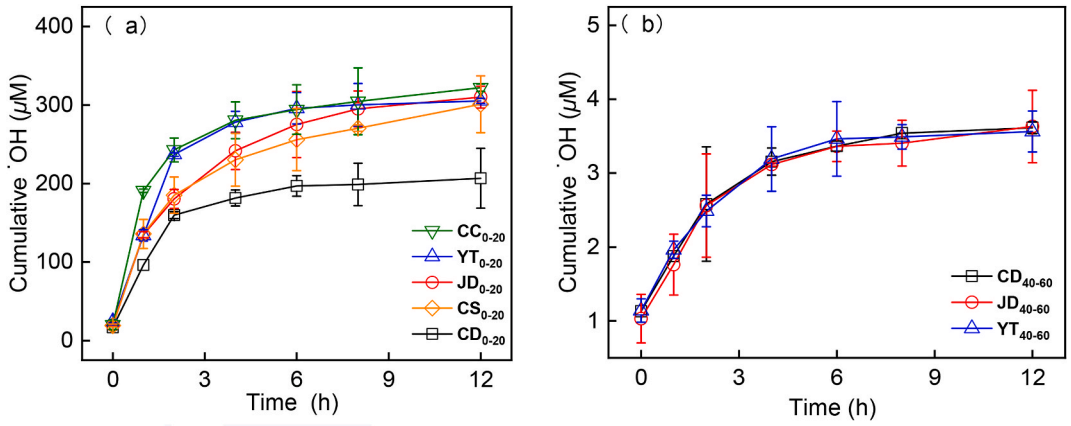


Fig. 1. The •OH production in the topsoil (a) and deepsoil (b) during the oxygenation processes.

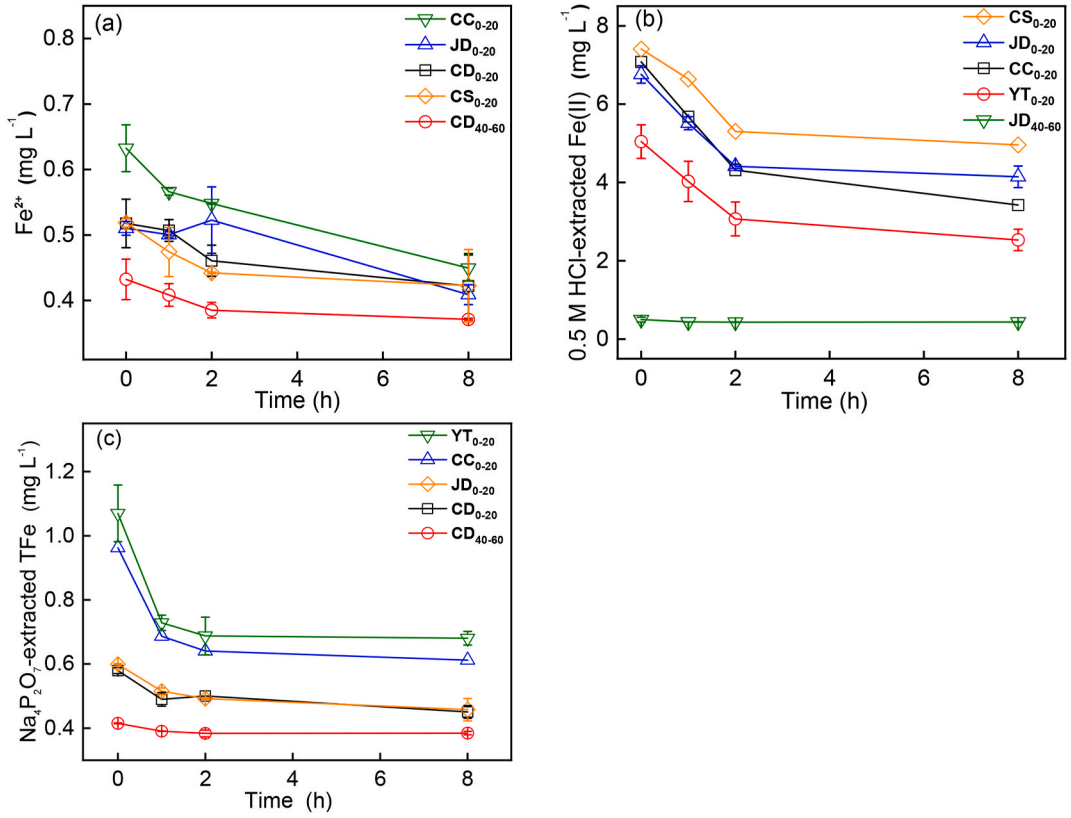


Fig. 2. Changes of aqueous Fe^{2+} (a), 0.5 M HCl-extracted Fe(II) (b), and $Na_4P_2O_7$-extracted TFe (c) during oxygenation processes.

slurries were 1.15 ± 0.034 mg L^{-1} h^{-1}, 0.99 ± 0.014 mg L^{-1} h^{-1}, 1.38 ± 0.008 mg L^{-1} h^{-1} and 1.05 ± 0.164 mg L^{-1} h^{-1}, respectively, which indicated the faster oxidation of 0.5 M HCl–Fe(II) occurred in the $CC_{0\text{-}20}$ slurry. After 8 h oxygenation, the oxidized content of 0.5 M HCl–Fe(II) in the $JD_{0\text{-}20}$, $YT_{0\text{-}20}$, $CC_{0\text{-}20}$ and $CS_{0\text{-}20}$ slurries were 2.61, 2.51, 3.65 and 2.45 mg L^{-1}, respectively, indicating more oxidation of 0.5 M HCl–Fe(II) was in $CC_{0\text{-}20}$ slurry than other paddy slurries. These results were also well corresponding to the aforementioned the higher •OH accumulation rate and the larger •OH accumulation in the $CC_{0\text{-}20}$ slurry after 8 h oxygenation than that of other paddy slurries. For different depths of soil slurries, the oxidized content of 0.5 M HCl–Fe(II) (2.34 mg L^{-1}) in the $JD_{0\text{-}20}$ slurry were much higher than that of $JD_{40\text{-}60}$ slurry (0.05 mg L^{-1}, Fig. 2b), which induced more •OH production after 8 h oxygenation. These combined results further suggested that •OH production was tightly related to 0.5 M HCl–Fe(II).

The initial content of $Na_4P_2O_7$-extracted TFe ($Na_4P_2O_7$-TFe) in the $JD_{0\text{-}20}$, $YT_{0\text{-}20}$, $CD_{0\text{-}20}$, $CD_{40\text{-}60}$ and $CC_{0\text{-}20}$ slurries were 0.60 ± 0.00, 1.06 ± 0.09, 0.58 ± 0.01, 0.41 ± 0.001 and 0.96 ± 0.04 mg L^{-1}, respectively (Fig. 2c). After 8 h oxygenation, the $Na_4P_2O_7$-TFe content in the $JD_{0\text{-}20}$, $YT_{0\text{-}20}$, $CD_{0\text{-}20}$, $CD_{40\text{-}60}$ and $CC_{0\text{-}20}$ slurries decreased to 0.46 ± 0.03, 0.68 ± 0.02, 0.45 ± 0.02, 0.38 ± 0.01 and 0.61 ± 0.00 mg L^{-1}, respectively. These results indicated that the $Na_4P_2O_7$-TFe also changed during oxygenation, which might partially participate the •OH production due to its complexation with organic matters (Garg et al., 2018; Jones et al., 2015; Rose and Waite, 2002). Comparing to the $JD_{0\text{-}20}$, $YT_{0\text{-}20}$, $CC_{0\text{-}20}$ and $CS_{0\text{-}20}$ slurries, large decrease of $Na_4P_2O_7$-TFe occurred in the $CC_{0\text{-}20}$ (0.35 ± mg L^{-1}) and $YT_{0\text{-}20}$ (0.38 mg L^{-1}) slurries than other paddy slurries, which was also corresponding to the highest accumulation of •OH (322 ± 4 μM •OH in the $CC_{0\text{-}20}$ slurry, 305 ± 3 μM •OH in the $YT_{0\text{-}20}$ slurry). Besides, the higher initial content (0.45 ± 0.02

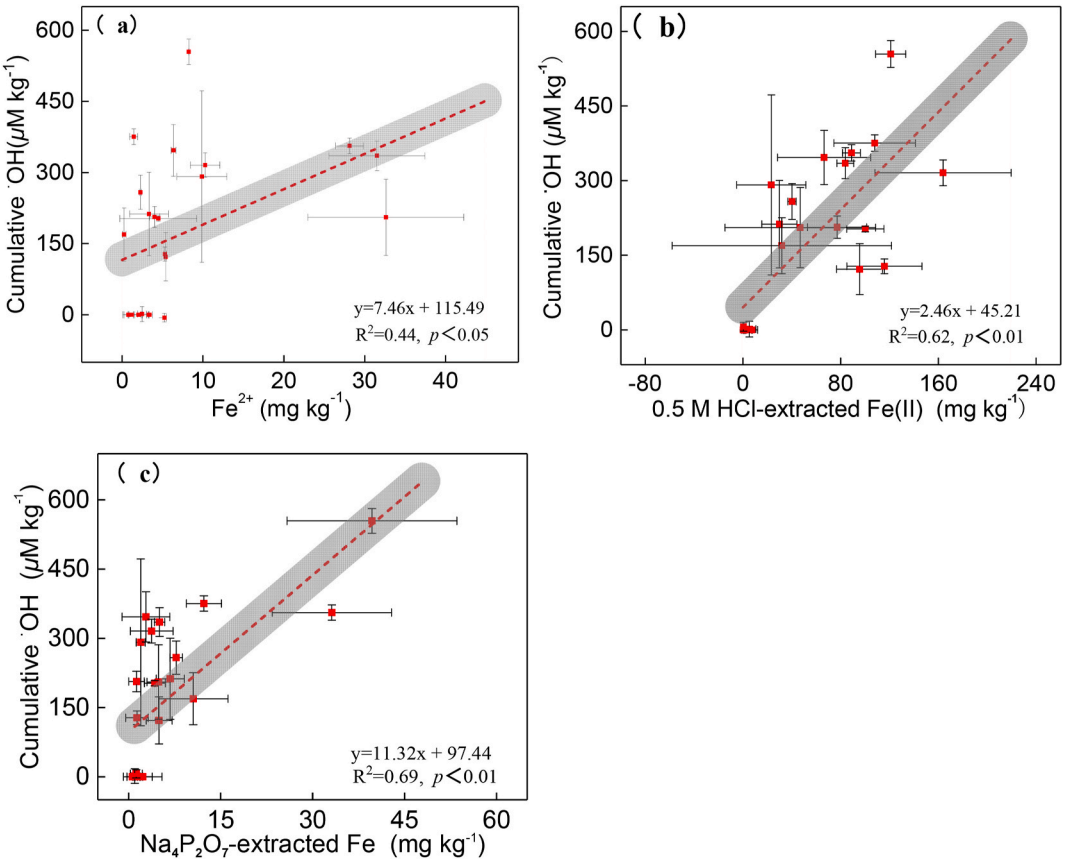


Fig. 3. Correlation analyses between •OH production and the content of Fe^{2+} (a), 0.5 M HCl-extracted Fe(II) (b), and $Na_4P_2O_7$-extracted TFe (c). The filled region represents the 95% confidence interval.

mg L^{-1}) of $Na_4P_2O_7$-TFe in the $CD_{0\text{-}20}$ slurry was also corresponded to more •OH formation than that of $CD_{40\text{-}60}$ slurry. Together, these results indicated that the $Na_4P_2O_7$-TFe may also contribute to •OH production during oxygenation.

3.3. Correlations analyses of active Fe species to •OH production

To further explore the relations between active Fe species and •OH production, the correlation analyses were further performed in these paddy slurries. Results showed that the Fe^{2+} ($R^2 = 0.439$, $p < 0.05$) and 0.5 M HCl–Fe(II) ($R^2 = 0.617$, $p < 0.01$) were both positively related with •OH production (Fig. 3a and b). But the oxidized content of 0.5 M HCl–Fe(II) (3.9–7.4 mg L^{-1}) was much higher than that of Fe^{2+} (0.40–1.85 mg L^{-1}), which indicated that 0.5 M HCl–Fe(II) contributed to more •OH production than that of aqueous Fe^{2+}. Besides, the $Na_4P_2O_7$-TFe was also positively ($R^2 = 0.687$, $p < 0.01$) correlated with the •OH accumulation (Fig. 3c), which indicated that $Na_4P_2O_7$–Fe might also contribute to •OH production in oxygenation of these slurries. Presumably, the contribution of $Na_4P_2O_7$–Fe to •OH production might be lower than that of 0.5 M HCl–Fe(II) due to the lower initial content of $Na_4P_2O_7$–Fe (0.58–1.06 mg L^{-1}) than that of 0.5 M HCl–Fe(II) (5.04–7.40 mg L^{-1}) in these paddy slurries.

3.4. Transformation of Fe species during oxygenation

The content of $Na_4P_2O_7$-TFe and NH_2OH·HCl-TFe decreased gradually during 8 h oxygenation (Fig. 2c and S4), indicating the transformation of these Fe species (Chen et al., 2018a; Thompson et al., 2006). In contrary, the $Na_2S_2O_4$-TFe slightly increased during oxygenation probably due to the accumulation of the transformed $Na_4P_2O_7$-TFe and NH_2OH·HCl-TFe species (Thompson et al., 2006). These results indicated that the organic-complexed Fe and low-crystalline Fe phases easily transformed during oxygenation. Our findings were also consistent with previous studies, which documented that low-crystalline Fe oxides (such as ferrihydrite) easily transformed into high-crystalline Fe oxides (e.g., hematite and goethite) during soil redox fluctuations (Coward et al., 2018; Thompson et al., 2006; Winkler et al., 2018).

3.5. Characterization of Fe species by XPS and mössbauer spectroscopy

The XPS spectra of Fe (2p3/2) before oxidation and after oxidation were examined to characterize the changes of near surface Fe in these paddy slurries, as XPS only detects the depth of a few nanometers. As presented in Fig. 4, surface Fe(II) content dramatically decreased from 15.0 to 11.2% after 8 h oxidation in the $CD_{0\text{-}20}$ slurry, which indicated the surface Fe(II) was quickly oxidized by O_2. Similar results were also observed in the $CS_{0\text{-}20}$ slurry (Fig. 4e and f). However, consistent surface Fe(II) content was observed in the $YT_{0\text{-}20}$ slurry before and after 8 h oxygenation, indicating the formed surface Fe(III) species was reduced during oxygenation. This was reasonable, because higher fraction of clay particles were observed in the $YT_{0\text{-}20}$ slurry than that of other paddy slurries (Table S1). Hence, the high content of interior Fe(II) entities (e.g., Fe(II)–Fe(II)–Fe(II) entities) in Fe-bearing clay minerals might induce the re-generation of surface Fe(II) via electrons transfer through basal siloxane plane or Fe(II)–O–Fe(III) linkages (Yuan et al., 2018). Besides, the surface Fe(II) of the $YT_{40\text{-}60}$ slurry decreased from 10.4 to 7.6% (Figure S5), which was different from that of the $YT_{0\text{-}20}$ slurry probably due to the lower clay fraction in the $CD_{40\text{-}60}$ slurry (Table S1). These results collectively illustrated that surface Fe(II) (i.e., surface-bound Fe (II) extracted by 0.5 M HCl) can be largely oxidized during oxygenation processes, which might be highly affected by the clay fractions in paddy slurries due to interior to exterior electron-transfer processes.

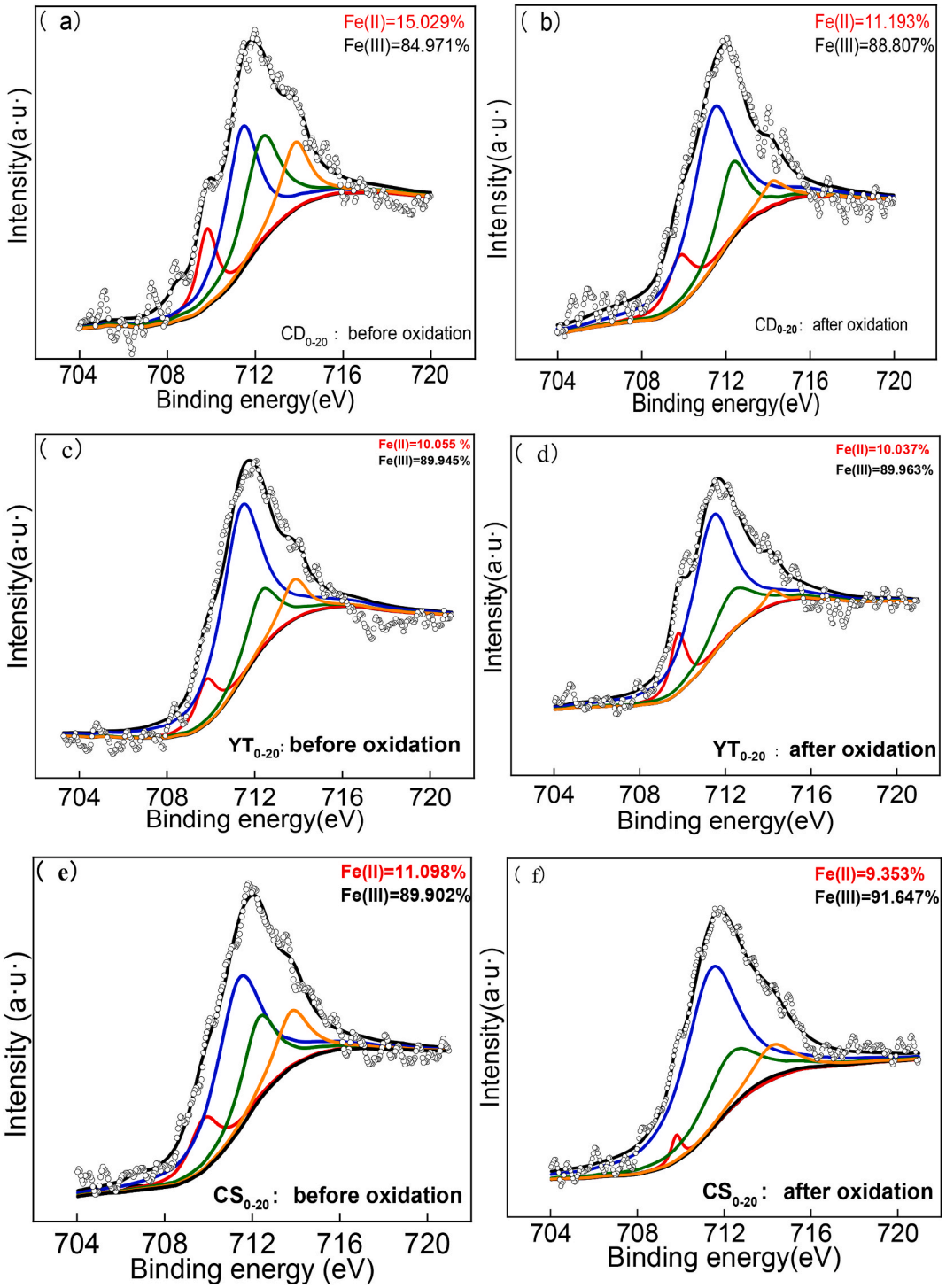


Fig. 4. The XPS spectra of $Fe2p_{3/2}$ for the $CD_{0\text{-}20}$ (a, b), $YT_{0\text{-}20}$ (c, d) and $CS_{0\text{-}20}$ (e, f) slurries before and after 8 h oxygenation.

Mössbauer spectra was also applied to characterize the changes of Fe species during oxygenation. Results showed that two doublets were mainly identified before and after oxygenation in the $CS_{0\text{-}20}$ slurry, which represented the Fe(II) in primary minerals, clays or sorbed, and Fe (III) in silicates and in the organic complexes, respectively (Fig. 5). After 8 h oxygenation, total Fe(II) decreased from 25.4 to 17.2%. Correspondingly, the Fe(III) phases increased from 74.6 to 82.8%. These results further confirmed that Fe(II) was oxygenated during oxygenation of anoxic paddy slurry, which might transform into high-crystalline minerals during oxygenation (Thompson et al., 2006; Winkler et al., 2018).

3.6. Mechanisms of Fe(II) species oxidation and $^{\bullet}OH$ formation

The above findings corroborated that 0.5 M HCl–Fe(II) (e.g., complexed or surface-bound Fe(II), and Fe(II) in low-crystalline minerals) mainly contributed to the $^{\bullet}OH$ production in paddy slurries, whereas the aqueous Fe^{2+} contributed to a much lesser extent. To investigate the contribution of aqueous Fe^{2+} and surface-exposed Fe(II) species in driving $^{\bullet}OH$ production, the BPY was added to screen the effects of these Fe(II) species due to its strong chelating ability (Katsoyiannis et al., 2008). The addition of 30 mM BPY suppressed 31% of $^{\bullet}OH$ production in the CC0-20 slurry, which might indicate that the oxidation of structural

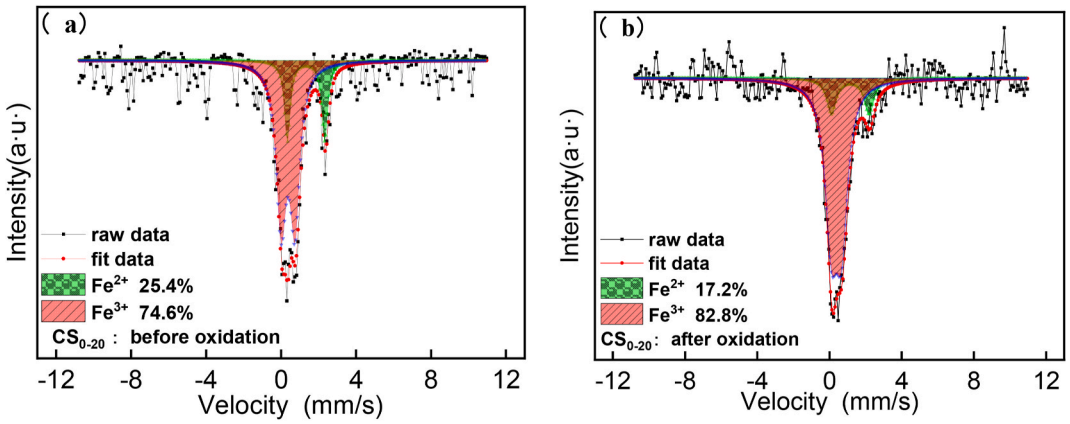


Fig. 5. The ^{57}Fe Mössbauer spectra (13 K) of the $CS_{0\text{-}20}$ slurry before (a) and after 8 h oxygenation (b).

Fe(II) species contributed to more •OH production than Fe^{2+} and surface Fe(II) species in the $CC_{0\text{-}20}$ slurry (Figure S6a) (Tong et al., 2016). The structural Fe(II) might include the Fe(II) species in low-crystalline minerals (e.g., siderite) as extracted by 0.5 M HCl, which might not be screened by BPY. Hence, the Fe(II) in low-crystalline minerals might also contribute to portion of •OH production. More studies should be employed to further investigate the underlined mechanisms in oxidation of these Fe(II) species.

The 0.5 M HCl–Fe(II) can dissolve the majority of complexed or surface-adsorbed Fe, and low-crystalline Fe species (e.g., ferrihydrite and siderite) (Chen et al., 2018a; Voelz et al., 2019). The components, such as Fe/Al-(hydr)oxides, silicates, organic matter, etc., can adsorb large amount of Fe(II) (Hofstetter et al., 2003; Sparks, 2003), via surface functional groups (≡XO-, X = Si, Fe, Al, H and C═O, etc.) (Xie et al., 2020). Oxidation of surface-adsorbed Fe(II) has been proposed to follow the inner- or out-sphere mechanisms depending on the pH, types of ligands and Fe(II) coordination sites (Nordin et al., 1998; Qian et al., 2019; Rosso and Morgan, 2002), which showed that the hydrolyzed forms of Fe(II) species (e.g., $Fe(OH_2)_5(OH)^+$) was oxidized by an inner-sphere pathway (Rosso and Morgan, 2002). Besides, it has also been proposed that electron-poor ligands decrease the Fe(II) reactivity but promoted the •OH production via outer-sphere mechanisms. While the electron-rich ligands enhances the Fe(II) reactivity and accelerates the production of other oxidants (e.g., Fe(IV)) via inner-sphere mechanisms during O_2-mediated oxygenation processes (Xie et al., 2020). Hence, the pH of $CS_{0\text{-}20}$ and $JD_{0\text{-}20}$ were higher than that of $YT_{0\text{-}20}$ slurry (Table S1), inducing faster oxidation of Fe(II) species via inner-sphere mechanisms (Fig. 2b) and consequently produced less •OH (Fig. 1a), while more Fe(II) species in the YT slurry was slowly oxidized through outer-sphere mechanisms and thus produced more •OH. In addition, the higher content of organic matter was observed in the $CC_{0\text{-}20}$ slurry, which indicated more •OH could be produced from the oxygenation of organic matter or organic ligands-enhanced Fe oxidation (Jiang et al., 2015; Jones et al., 2015; Page et al., 2012). Although the proposed mechanisms can roughly interpret our results, more experimental and theoretical proofs are needed to validate its applicability in future studies.

To further investigate the electron transfer pathways of •OH formation, we carried out quenching experiments with addition of catalase and NBT to screen the effects of $O_2^{\cdot-}$ and H_2O_2 (Figure S6) (Liao et al., 2019; Tong et al., 2016). After quenching H_2O_2 and $O_2^{\cdot-}$, the cumulative •OH significantly decreased by 22–30% in the $CC_{0\text{-}20}$ slurry, which indicated one- and two-electron transfer mechanisms were both responsible for the •OH production (Tong et al., 2016). The two-electron transferring mechanisms widely occur in the oxygenation of OM-complexed Fe(II) and FeS (Cheng et al., 2016, 2020; Lee et al., 2014). These results probably suggested that the oxygenation of active Fe(II) species was through the following equations (Eqs. (1)–(4)) based on the Haber-Weiss mechanisms:

$$\equiv FeOH^+ + O_2 + H^+ \rightarrow FeOH^{2+} + HO_2^{\cdot} \quad (1)$$

$$\equiv Fe^{2+} + HO_2^{\cdot} + H_2O \rightarrow FeOH^{2+} + H_2O_2 \quad (2)$$

$$\equiv Fe^{2+} + H_2O_2 \rightarrow FeOH^{2+} + {}^{\cdot}OH \quad (3)$$

$$\equiv Fe^{2+} + {}^{\cdot}OH \rightarrow Fe^{3+} + OH^- \quad (4)$$

3.7. Effects of •OH on the IMP degradation and OC transformation

The IMP was relatively stable under anoxic conditions in the $CC_{0\text{-}20}$ slurry, with negligible changes (<2%) observed after 12 h reaction. However, IMP concentration decreased by 25% after 12 h oxygenation (Fig. 6a). Besides, IMP concentration rarely decreased with the addition of 10% methanol, which indicated the •OH was responsible for IMP degradation. The oxidation pathway of IMP by •OH oxidation was proposed in Fig. 6b. Based on the results of HPLC-MS/MS, degradation products such as the IMP-urea, IMP-olefin, 6-chloronicotinic acid were identified in current study (Figure S7) (Bonmatin et al., 2003; Bourgin et al., 2011; Zhou et al., 2018). It has been proposed that the IMP was mainly attacked by •OH occurred in the C^1–N bond of tertiary amines (Route 1) and *N*-dealkylation of amines on the imidazolidine ring (Route 2) (Chen et al., 2019b). Besides, hydroxylation may occur on each carbon atom at α position of amine group, producing the mono-hydroxylated IMP (Route 5). Hydroxylation possibly occurred on each α-position carbon atom of the amine group and thus induced the production of mono-hydroxy IMP, which can be further oxidized to a carbonyl group or another α-position carbon atom by the second hydroxylation reaction or both further degraded to form mono-hydroxy IMP. The cleavage of the bond between *N*–NO_2 and imine (NH =) groups was followed by oxidative cleavage of the C–N bond between the oxidized imidazolidine ring and the pyridine ring, as shown in Route 3 (Kitsiou et al., 2009; Li et al., 2019). In addition, IMP-urea can be produced via the •OH-attacking reactions (Route 3) (Bourgin et al., 2011; Zhang et al., 2018b).

To examine the effects of •OH on the oxidation of organic carbon, XPS was used to characterize the variations of functional groups of OC before and after oxygenation. Results showed that the contents of the C═C, C–O and C═O groups in the $CS_{0\text{-}20}$ slurry were 42.04%, 7.87%, and 5.96% before the oxidation reaction, respectively (Figure S8k, l). After 8 h oxidation, the content of these species decreased to 24.75%, 4.12% and 3.44%, respectively. Meanwhile, the contents of the C–H, C–C and COO^- groups increased from 25.51%, 15.72%, and 2.89%–40.20%, 22.92%, and 4.57%, respectively. These results suggested that •OH oxidation might largely induce the oxidative transformation of OC (Waggoner et al., 2015; Zeng et al., 2020). The •OH induced addition of

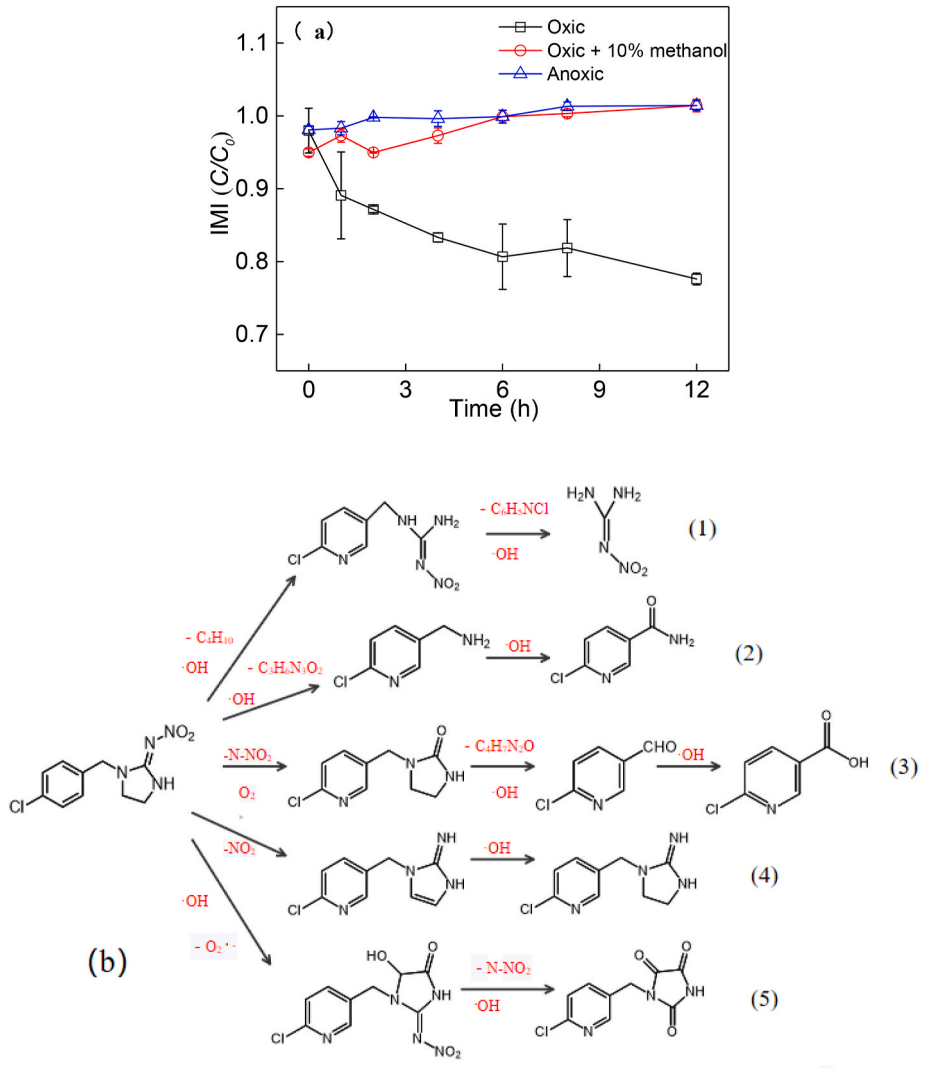


Fig. 6. The IMP degradation during oxygenation processes (a) and proposed degradation pathways of IMP by •OH (b).

COO^- to aromatic structures usually involves a ring-opening scheme, which includes depolymerization and demethylation due to cleavage of the aromatic ring, often producing unsaturated aliphatic and hydroxylated carboxylic acid-containing structures (Waggoner et al., 2015). It suggests that multiple reaction pathways might have occurred simultaneously during oxygenation processes, including both the hydrogenation and proton abstraction (Zeng et al., 2020).

In FTIR spectra of $CD_{0\text{-}20}$ slurry (Table S3 and Figure S9), the most prominent bands before oxidation were the C–O stretch on C–O–C at 1081 cm^{-1}, C═O signal enhancement on quinone ring at 1397 cm^{-1}, C═O signal on quinone ring at 1630 cm^{-1}, symmetric and asymmetric vibrations on CH_3 and CH_2 at 2859 cm^{-1} and 2926 cm^{-1}, O–H signal at 3438, 3822 and 3699 cm^{-1}, respectively (Cheng et al., 2006; Tatzber et al., 2007). After 8 h oxygenation, the above functional groups largely changed. The newly formed band of aromatic carboxyl groups appeared at 916 cm^{-1}, suggesting more carboxyl groups were produced during oxygenation processes (Waggoner et al., 2015; Zeng et al., 2020). Besides, new stretching band of C–O–C appeared at 1079 cm^{-1}, which was accompanied with the enhancement of the asymmetric COO^- band at 1595 cm^{-1} and the symmetric COO^- band at 1397 cm^{-1}. The C═O signal on quinone ring at 1397 cm^{-1}, symmetric C–H and asymmetric C–H tensile vibrations of $–CH_3$/$–CH_2–$ at 2859/2926 cm^{-1} became stronger after oxygenation (Table S3 and Figure S9). These collective results probably suggested that •OH induced OC oxidation can produce more C═O, COO^- and C–H functional groups.

4. Conclusions

This study investigated the active Fe(II) species mediating the •OH production and organic contaminant degradation during oxygenation of anoxic paddy slurries. Our findings showed that extensive •OH production can be produced during oxygenation of anoxic paddy slurries, but it was discrepant in different paddy slurries (40.3–1061.4 μmol kg^{-1}). Besides, the 0.5 M HCl–Fe(II) played an important role in driving •OH formation based on the wet-chemical extraction analyses, which was largely ignored in the previous studies. The IMP could be efficiently degraded by •OH, which also produced the IMP-urea, IMP-olefin and 6-chloronicontinic acid via •OH-attacking mechanisms. The •OH oxidation can induce the oxidative transformation of OC, which was underpinned by the increasing portion of the C–H, C–C, and COO groups after oxygenation. These findings may provide the potential strategies for controlling organic pollution in paddy fields via increasing the frequencies of redox fluctuations and improving the soil aeration, etc. Our findings improve the understanding of the Fe phases-mediated reactive oxygen species formation (e.g., •OH) and associated organic contaminants transformation during soil redox fluctuations in agricultural fields.

Credit author statement

Wenchao Wang: Methodology, Writing - original draft. Danyu Huang: Writing - review & editing. Dixiang Wang: Writing - Methodology. Mengxi Tan: Writing – Methodology. Mengyuan Geng: Writing - review & editing. Changyin Zhu: Writing - review & editing. Ning Chen: Methodology, Writing - review & editing, Funding acquisition. Dongmei Zhou: Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by grants from the National Key R&D Program of China (2019YFC1805704) and the Natural Science Foundation of Jiangsu Province (BK20200323).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemosphere.2021.131565.

References

Amonette, J.E., Tampleton, J.C., 1998. Improvements to the quantitative assay of nonrefractory minerals for Fe(II) and total Fe using 1,10-phenanthroline. Clay Clay Miner. 46 (1), 51–62.
Bonmatin, J.M., Moineau, I., Charvet, R., Fleche, C., Colin, M.E., Bengsch, E.R., 2003. A LC/APCI-MS/MS method for analysis of imidacloprid in soils, in plants, and in pollens. Anal. Chem. 75 (9), 2027–2033.
Bourgin, M., Violleau, F., Debrauwer, L., Albet, J., 2011. Ozonation of imidacloprid in aqueous solutions: reaction monitoring and identification of degradation products. J. Hazard Mater. 190 (1), 60–68.
Chen, C., Meile, C., Wilmoth, J., Barcellos, D., Thompson, A., 2018a. Influence of pO_2 on iron redox cycling and anaerobic organic carbon mineralization in a humid tropical forest soil. Environ. Sci. Technol. 52 (14), 7709–7719.
Chen, N., Fang, G., Liu, G., Zhou, D., Gao, J., Gu, C., 2018b. The effects of Fe-bearing smectite clays on •OH formation and diethyl phthalate degradation with polyphenols and H2O2. J. Hazard Mater. 357 (5), 483–490.
Chen, N., Fang, G., Liu, G., Zhou, D., Gao, J., Gu, C., 2019a. The degradation of diethyl phthalate by reduced smectite clays and dissolved oxygen. Chem. Eng. J. 355 (1), 247–254.
Chen, S., Deng, J., Deng, Y., Gao, N., 2019b. Influencing factors and kinetic studies of imidacloprid degradation by ozonation. Environ. Technol. 40 (16), 2127–2134.
Cheng, C.-H., Lehmann, J., Thies, J.E., Burton, S.D., Engelhard, M.H., 2006. Oxidation of black carbon by biotic and abiotic processes. Org. Geochem. 37 (11), 1477–1488.
Cheng, D., Neumann, A., Yuan, S., Liao, W., Qian, A., 2020. Oxidative degradation of organic contaminants by FeS in the presence of O_2. Environ. Sci. Technol. 54 (7), 4091–4101.
Cheng, D., Yuan, S., Liao, P., Zhang, P., 2016. Oxidizing impact induced by mackinawite (FeS) nanoparticles at oxic conditions due to production of hydroxyl radicals. Environ. Sci. Technol. 50 (21), 11646–11653.
Coward, E.K., Thompson, A., Plante, A.F., 2018. Contrasting Fe speciation in two humid forest soils: insight into organomineral associations in redox-active environments. Geochim.Cosmochim. Acta 238, 68–84.
Garg, S., Jiang, C., Waite, T.D., 2018. Impact of pH on iron redox transformations in simulated freshwaters containing natural organic matter. Environ. Sci. Technol. 52 (22), 13184–13194.
Gligorovski, S., Strekowski, R., Barbati, S., Vione, D., 2015. Environmental implications of hydroxyl radicals (•OH). Chem. Rev. 115 (24), 13051–13092.
Han, R., Lv, J., Huang, Z., Zhang, S., Zhang, S., 2020. Pathway for the production of hydroxyl radicals during the microbially mediated redox transformation of iron (oxyhydr)oxides. Environ. Sci. Technol. 54 (2), 902–910.
Hanke, A., Cerli, C., Muhr, J., Borken, W., Kalbitz, K., 2013. Redox control on carbon mineralization and dissolved organic matter along a chronosequence of paddy soils. Eur. J. Soil Sci. 64 (4), 476–487.
Heron, G., Crouzet, C., Bourg, A.C.M., Christensen, T.H., 1994. Speciation of Fe(II) and Fe(III) in contaminated aquifer sediments using chemical extraction techniques. Environ. Sci. Technol. 28 (9), 1698–1705.
Hofstetter, T.B., Schwarzenbach, R.P., Haderlein, S.B., 2003. Reactivity of Fe(II) species associated with clay minerals. Environ. Sci. Technol. 37 (3), 519–528.
Jiang, C., Garg, S., Waite, T.D., 2015. Hydroquinone-mediated redox cycling of iron and concomitant oxidation of hydroquinone in oxic waters under acidic conditions: comparison with iron–natural organic matter interactions. Environ. Sci. Technol. 49 (24), 14076–14084.
Jones, A.M., Griffin, P.J., Waite, T.D., 2015. Ferrous iron oxidation by molecular oxygen under acidic conditions: the effect of citrate, EDTA and fulvic acid. Geochem. Cosmochim. Acta 160, 117–131.
Katsoyiannis, I.A., Ruettimann, T., Hug, S.J., 2008. pH dependence of Fenton reagent generation and As(III) oxidation and removal by corrosion of zero valent iron in aerated water. Environ. Sci. Technol. 42 (19), 7424–7430.
Kitsiou, V., Filippidis, N., Mantzavinos, D., Poulios, I., 2009. Heterogeneous and homogeneous photocatalytic degradation of the insecticide imidacloprid in aqueous solutions. Appl. Catal. B Environ. 86 (1), 27–35.
Kögel knabner, I., Amelung, W., Cao, Z., Fiedler, S., Frenzel, P., Jahn, R., Kalbitz, K., Kölbl, A., Schloter, M., 2010. Biogeochemistry of paddy soils. Geoderma 157 (1), 1–14.
Lee, J., Kim, J., Choi, W., 2014. Oxidation of aquatic pollutants by ferrous–oxalate complexes under dark aerobic conditions. J. Hazard Mater. 274, 79–86.
Lesueur, C., Gartner, M., Mentler, A., Fuerhacker, M., 2008. Comparison of four extraction methods for the analysis of 24 pesticides in soil samples with gas chromatography–mass spectrometry and liquid chromatography-ion trap-mass spectrometry. Talanta 75 (1), 284–293.
Li, Y., Long, L., Ge, J., Li, H., Zhang, M., Wan, Q., Yu, X., 2019. Effect of imidacloprid uptake from contaminated soils on vegetable growth. Jounal of Agricultural Food Chemistry 67 (26), 7232–7242.
Liao, P., Yu, K., Lu, Y., Wang, P., Liang, Y., Shi, Z., 2019. Extensive dark production of hydroxyl radicals from oxygenation of polluted river sediments. Chem. Eng. J. 368, 700–709.
Liu, X., Yuan, S., Tong, M., Liu, D., 2017. Oxidation of trichloroethylene by the hydroxyl radicals produced from oxygenation of reduced nontronite. Water Res. 113, 72–79.
Mopper, K., Zhou, X.L., 1990. Hydroxyl radical photoproduction in the sea and its potential impact on marine processes. Science 250 (4981), 661–664.
Nordin, J.P., Sullivan, D.J., Phillips, B.L., Casey, W.H., 1998. AnO^{17}-NMR study of the exchange of water on $AlOH(H_2O)_5^{2+(aq)}$. Inorg. Chem. 37, 4760–4763.
Page, S.E., Kling, G.W., Sander, M., Harrold, K.H., Logan, J.R., McNeill, K., Cory, R.M., 2013. Dark formation of hydroxyl radical in Arctic soil and surface waters. Environ. Sci. Technol. 47 (22), 12860–12867.
Page, S.E., Sander, M., Arnold, W.A., Mcneill, K., 2012. Hydroxyl radical formation upon oxidation of reduced humic acids by oxygen in the dark. Environ. Sci. Technol. 46 (3), 1590–1597.
Poulton, S.W., Canfield, D.E., 2005. Development of a sequential extraction procedure for iron: implications for iron partitioning in continentally derived particulates. Chem. Geol. 214 (3), 209–221.
Qian, A., Yuan, S., Xie, S., Tong, M., Zhang, P., Zheng, Y., 2019. Oxidizing capacity of iron electrocoagulation systems for refractory organic contaminants transformation. Environ. Sci. Technol. 53, 12629–12638.
Qin, W., Wang, Y., Fang, G., Liu, C., Sui, Y., Zhou, D., 2016. Oxidation mechanism of As (III) in the presence of polyphenols: new insights into the reactive oxygen species. Chem. Eng. J. 285, 69–76.
Rose, A.L., Waite, T.D., 2002. Kinetic model for fe(II) oxidation in seawater in the absence and presence of natural organic matter. Environ. Sci. Technol. 36 (3), 433–444.
Rosso, K.M., Morgan, J.J., 2002. Outer-sphere electron transfer kinetics of metal ion oxidation by molecular oxygen. Geochem. Cosmochim. Acta 66 (24), 4223–4233.
Santana-Casiano, J.M., González-Dávila, M., Millero, F.J., 2005. Oxidation of nanomolar levels of fe(II) with oxygen in natural waters. Environ. Sci. Technol. 39 (7), 2073–2079.
Sharma, S., Singh, B., 2014. Metabolism and persistence of imidacloprid in different types of soils under laboratory conditions. Int. J. Environ. Anal. Chem. 94, 1100–1112.
Sparks, D.L., 2003. Environmental Soil Chemistry. Elsevier Science, USA).
Tatzber, M., Stemmer, M., Spiegel, H., Katzlberger, C., Haberhauer, G., Mentler, A., Gerzabek, M.H., 2007. FTIR-spectroscopic characterization of humic acids and humin fractions obtained by advanced NaOH, $Na_4P_2O_7$, and Na_2CO_3 extraction procedures. J. Plant Nutr. Soil Sci. 170 (4), 522–529.
Thompson, A., Chadwick, O.A., Rancourt, D.G., Chorover, J., 2006. Iron-oxide crystallinity increases during soil redox oscillations. Geochem. Cosmochim. Acta 70 (7), 1710–1727.
Tong, M., Yuan, S., Ma, S., Jin, M., Liu, D., Cheng, D., Liu, X., Gan, Y., Wang, Y., 2016. Production of abundant hydroxyl radicals from oxygenation of subsurface sediments. Environ. Sci. Technol. 50 (1), 214–221.
Voelz, J.L., Johnson, N.W., Chun, C.L., Arnold, W.A., Penn, R.L., 2019. Quantitative dissolution of environmentally accessible iron residing in iron-rich minerals: a Review. ACS Earth Space Chemistry 3 (8), 1371–1392.
Vogelsang, V., Kaiser, K., Wagner, F.E., Jahn, R., Fiedler, S., 2016. Transformation of clay-sized minerals in soils exposed to prolonged regular alternation of redox conditions. Geoderma 278, 40–48.
Waggoner, D.C., Chen, H., Willoughby, A.S., Hatcher, P.G., 2015. Formation of black carbon-like and alicyclic aliphatic compounds by hydroxyl radical initiated degradation of lignin. Org. Geochem. 82, 69–76.
Wang, D., Huang, D., Wu, S., Fang, G., Zhu, F., Chen, N., Liu, S., Zhu, C., Zhou, D., 2021. Pyrogenic carbon initiated the generation of hydroxyl radicals from the oxidation of sulfide. Environ. Sci. Technol. 55, 6001–6011.
Wang, J., Wang, P., Gu, Y., Kopittke, P.M., Zhao, F., Wang, P., 2019a. Iron–manganese (oxyhydro)oxides, rather than oxidation of sulfides, determine mobilization of Cd during soil drainage in paddy soil systems. Environ. Sci. Technol. 53 (5), 2500–2508.
Wang, Q., Rao, P., Li, G., Dong, L., Zhang, X., Shao, Y., Gao, N., Chu, W., Xu, B., An, N., Deng, J., 2020. Degradation of imidacloprid by UV-activated persulfate and

peroxymonosulfate processes: kinetics, impact of key factors and degradation pathway. Ecotoxicol. Environ. Saf. 187, 109779–109787.
Wang, X., Liu, B., Ma, J., Zhang, Y., Hu, T., Zhang, H., Feng, Y., Pan, H., Xu, Z., Liu, G., Lin, X., Zhu, J., Bei, Q., Xie, Z., 2019b. Soil aluminum oxides determine biological nitrogen fixation and diazotrophic communities across major types of paddy soils in China. Soil Biol. Biochem. 131, 81–89.
Watanabe, T., Katayanagi, N., Agbisit, R., Llorca, L., Hosen, Y., Asakawa, S., 2021. Influence of alternate wetting and drying water-saving irrigation practice on the dynamics of Gallionella-related iron-oxidizing bacterial community in paddy field soil. Soil Biol. Biochem. 152, 108064–108074.
Williams, A.G.B., Scherer, M.M., 2001. Kinetics of Cr(VI) reduction by carbonate green rust. Environ. Sci. Technol. 35 (17), 3488–3494.
Winkler, P., Kaiser, K., Thompson, A., Kalbitz, K., Fiedler, S., Jahn, R., 2018. Contrasting evolution of iron phase composition in soils exposed to redox fluctuations. Geochem. Cosmochim. Acta 235, 89–102.
Xie, W., Yuan, S., Tong, M., Ma, S., Liao, W., Zhang, N., Chen, C., 2020. Contaminant degradation by ·OH during sediment oxygenation: dependence on Fe(II) species. Environ. Sci. Technol. 54 (5), 2975–2984.
Xu, J., Li, X., Sun, G., Cui, L., Ding, L., He, C., Li, L., Shi, Q., Smets, B.F., Zhu, Y., 2019. Fate of labile organic carbon in paddy soil is regulated by microbial ferric iron reduction. Environ. Sci. Technol. 53 (15), 8533–8542.
Yu, G., Kuzyakov, Y., 2021. Fenton chemistry and reactive oxygen species in soil: abiotic mechanisms of biotic processes, controls and consequences for carbon and nutrient cycling. Earth Sci. Rev. 214, 103525–103538.
Yuan, S., Liu, X., Liao, W., Zhang, P., Wang, X., Tong, M., 2018. Mechanisms of electron transfer from structural Fe(II) in reduced nontronite to oxygen for production of hydroxyl radicals. Geochem. Cosmochim. Acta 223, 422–436.
Zeng, Q., Wang, X., Liu, X., Huang, L., Hu, J., Chu, R.K., Tolič, N., Dong, H., 2020. Mutual interactions between reduced Fe-bearing clay minerals and humic acids under dark, oxygenated condition: hydroxyl radical generation and humic acid transformation. Environ. Sci. Technol. 54, 15013–15023.
Zhang, P., Huang, W., Ji, Z., Zhou, C., Yuan, S., 2018a. Mechanisms of hydroxyl radicals production from pyrite oxidation by hydrogen peroxide: surface versus aqueous reactions. Geochem. Cosmochim. Acta 238, 394–410.
Zhang, P., Ren, C., Sun, H., Min, L., 2018b. Sorption, desorption and degradation of neonicotinoids in four agricultural soils and their effects on soil microorganisms. Sci. Total Environ. 615, 59–69.
Zhang, P., Yuan, S., 2017. Production of hydroxyl radicals from abiotic oxidation of pyrite by oxygen under circumneutral conditions in the presence of low-molecular-weight organic acids. Geochem. Cosmochim. Acta 218, 153–166.
Zhang, P., Yuan, S., Chen, R., Bu, X., Tong, M., Huang, Q., 2020. Oxygenation of acid sulfate soils stimulates CO_2 emission: roles of acidic dissolution and hydroxyl radical oxidation. Chem. Geol. 533, 119437–119449.
Zhao, L., Fang, Q., Algeo, T.J., Lu, A., Yin, K., Duan, Z., Hong, H., 2021. Formation of plinthite mediated by redox fluctuations and chemical weathering intensity in a Quaternary red soil, southern China. Geoderma 386, 114924–114937.
Zhou, X., Tian, Y., Liu, X., Huang, L., Wen, Y., 2018. Reduction of imidacloprid by sponge iron and identification of its degradation products. Water Environ. Res.: A Research Publication of the Water Environment Federation 90 (12), 2049–2055.